

1. Functional derivatives: For a three dimensional elastic medium, the potential energy is

$$V = \frac{\tau}{2} \int d^3x (\nabla\phi)^2,$$

and the kinetic energy is

$$T = \frac{\rho}{2} \int d^3x \left(\frac{\partial\phi}{\partial t} \right)^2.$$

Use the variational principle to show that ϕ obeys the wave equation

$$\nabla^2\phi = \frac{1}{v^2} \frac{\partial^2\phi}{\partial t^2}$$

where v is the wave velocity, ρ the density of the medium and τ the tension.

2. Functional derivatives: Show that if

$$Z[J] = \exp \left[-\frac{1}{2} \int d^4x d^4y J(x) \Delta(x-y) J(y) \right],$$

where $\Delta(x) = \Delta(-x)$, then

$$\frac{\delta Z[J]}{\delta J(z_0)} = -Z[J] \int d^4y \Delta(z_0 - y) J(y)$$

3. Derive the asymptotic forms of the correlation function $G(r)$ for (i) $r \gg \xi$, and (ii) $r \ll \xi$ with the prefactors. The conditions for cases (i) and (ii) correspond to $T \neq T_c$ and $T = T_c$ respectively. [Use the Fourier transform method as discussed, for example, in David Tong's notes or in Chowdhury-Stauffer].
4. Now solve for $G(r)$ in three dimensions by evaluating the contour integral. The Fourier transform

$$G(\vec{r}) = \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} d^3k e^{i\vec{k}\cdot\vec{r}} \tilde{G}(\vec{k}),$$

and

$$\tilde{G}(\vec{k}) = \frac{1}{k^2 + \xi^{-2}}.$$

What is $G(\vec{r})$ when $\xi \rightarrow \infty$?

5. Consider the Landau free energy

$$L = \int d^d x \left[\frac{1}{2} (\nabla\phi)^2 + \frac{1}{2} r_0 \phi^2 + \frac{1}{n!} u_0 \phi^n \right]$$

Use the Ginzburg criterion to find the upper critical dimension.
(Assume n is even.)